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People Are More Variable Than Their Hop Test Would Suggest: Hop Performance and Self-Reported Outcomes Over 11 Years Following ACL Reconstruction

Michael A. Girdwood[1] | Kay M. Crossley[1] | Brooke E. Patterson[1] | Ebonie K. Rio[1,2,3] | Timothy S. Whitehead[4] | Hayden G. Morris[5] | Adam G. Culvenor[1]

[1]La Trobe Sport and Exercise Medicine Research Centre, La Trobe University, Bundoora, Victoria, Australia | [2]The Australian Ballet, Melbourne, Victoria, Australia | [3]The Victorian Institute of Sport, Melbourne, Victoria, Australia | [4]OrthoSport Institute Victoria, Melbourne, Victoria, Australia | [5]Park Clinic Orthopaedics, Melbourne, Victoria, Australia

Correspondence: Adam G. Culvenor (a.culvenor@latrobe.edu.au)

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Keywords: anterior cruciate ligament | hop test | knee | rehabilitation

ABSTRACT

We aimed to report the trajectory of self-reported outcomes up to 11 years post-ACLR. We also explored the relationship between hop performance at 1 year and: (i) future self-reported knee outcomes; and (ii) risk of subsequent knee events. 124 participants (43 women, mean age 31 ± 8 years) were recruited at 1 year following hamstring-autograft ACLR. Hop performance was assessed with single-forward and side-hop tests. Follow-up was completed at 3 ($n = 114$), 5 ($n = 89$) and 11 years ($n = 72$) post-ACLR. Self-reported outcomes were assessed at each follow-up with the Knee injury Osteoarthritis Outcome Score (KOOS) pain and quality of life (QOL) subscales. Generalized linear mixed models estimated the relationship between hop performance and self-reported outcomes. Subsequent knee events (new injury/surgery) to either knee were recorded, with the relationship between hop performance and risk of subsequent knee events analyzed with Cox proportional hazards. Self-reported knee outcomes were stable (mean change < 10 points) across all timepoints but with major within-sample variability. There was a modest relationship between greater hop performance at 1 year and better future KOOS-pain (average marginal effect [AME] % improvement with + 1 cm single forward hop = 0.06% [95% CI 0.02–0.10]). A nonlinear spline relationship showed better single-forward hop performance was associated with better KOOS-QOL for scores < 108 cm, not present for higher hop scores > 108 cm. There were 21 index and 11 contralateral subsequent knee events. Hop performance was not related to risk of a subsequent knee event (hazard ratio index knee 0.99 [95% CI 0.98–1.02]). In conclusion, self-reported knee pain and quality of life were generally stable across the 11-year follow-up period. Greater hop performance at 1-year post-ACLR was related to better self-reported knee outcomes up to 11-year follow-up (of questionable clinical importance), but not associated with the risk of subsequent knee injury/surgery.

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1 | Introduction

For people suffering an anterior cruciate ligament (ACL) injury, the future can often be an "unknown horizon." [1] It can lead to changes in lifestyle, leisure, sport, and working arrangements. As many as 45% of people do not return to competitive sport [2], and 9%–39% of people will have major pain and symptoms 6-year post-ACLR [3]. The devastating consequences of ACL injuries, often occurring in early adulthood, have lifelong impacts—the risk of osteoarthritis is 4–6 fold greater [4], and the risk of total knee arthroplasty is seven times higher compared to uninjured populations [5]. Rehabilitation focuses on resolving key impairments (e.g., quadriceps strength deficits) to facilitate return-to-sport and activity and reduce the risk of future knee symptoms. It is also important for clinicians to educate patients on their prognosis and expectations after ACL injury to help mitigate future burden and maximize recovery. Forecasting prognosis accurately has been a quest for scientists and clinicians—we endeavor to advise and educate patients on what to expect and how to treat it. Yet modifiable prognostic factors after ACL injury remain elusive [6], as for many conditions.

Hop tests are commonly utilized as clinical prognostic tools after ACL reconstruction (ACLR)—they guide rehabilitation progress [7] and help to advise safe return-to-sport and activity [8]. Hop tests can provide insights into functional performance (e.g., quantitively as distance) and also movement quality (e.g., motor control and biomechanics). Thanks to their easy administration, most (95%) clinicians use them as criteria for return-to-sport [9], with many variations reported, and a battery recommended by guidelines [10]. A recent review of 10 studies showed greater forward hop performance was related to better self-reported symptoms and higher likelihood of return-to-sport, but was limited by very-low certainty evidence [11]. Another review ($n = 10$ studies) with similar findings, suggested relationships between hop performance and self-reported knee symptoms may be modest and clinically irrelevant [12]. Most studies in these reviews only followed people for ≤3 years [11, 12], which may not provide sufficient insight for patients into their long-term prognosis.

At more than 10 years after ACLR, people report persistent and often worsening knee-related quality of life (QOL) and pain [13, 14], compared with population norms [15]. Long-term prognostic studies measuring potential explanatory factors for self-reported knee outcome are rare [16, 17]. Understanding what influences ongoing knee health and function is critical as people transition into different lifestyle and activity habits, with additional work and parental responsibilities [1, 18]. Currently, there are no known modifiable risk factors for persistent symptoms or osteoarthritis after traumatic knee injuries [6]. A better understanding of the factors related to poorer outcomes after ACLR is critical to designing future interventions to manage the unacceptably high burden of knee problems for many people after ACLR. The extent to which modifiable factors such as hop performance might relate to long-term prognosis remains unclear. Therefore, the aims of this study were to report the trajectory of self-reported knee outcomes up to 11-year post-ACLR. We also aimed to explore the relationship between hop performance at 1-year post-ACLR and: (i) future self-reported knee outcomes up to 11-year post-ACLR; and (ii) risk of subsequent knee events.

2 | Methods

2.1 | Study Design

Previously recruited participants from a longitudinal cohort study were eligible for the current study [19]. Follow-up timepoints at 1-, 3-, and 5-year post-ACLR have been published [20–23], with detailed inclusion criteria for each timepoint detailed in a Supporting Information. Briefly, we consecutively recruited 124 participants 1 year after primary ACLR with a single-bundle hamstring–tendon autograft, aged between 18 and 50 years at the time of surgery from two Melbourne-based surgeons. We invited all original participants (regardless of previous loss at any follow-up), to participate in an 11-year follow-up (originally planned for a 10-year follow-up, but affected by COVID-19-related delays). For the present study, we did not restrict inclusion based on surgical delay, or on injuries sustained since baseline recruitment (e.g., index re-ruptures or new contralateral ACL injuries). Ethical approval for this study was obtained from La Trobe University (HEC20529), and all participants provided informed consent.

2.2 | Procedures

At 11-year post-ACLR, participants were invited to complete: (i) online self-reported outcomes (Knee injury and Osteoarthritis Outcome Score (KOOS)); and (ii) brief consultation (phone or in-person) with an experienced physiotherapist (9 years) to record knee injury and other medical history.

2.3 | Outcomes

2.3.1 | Prognostic Factors: Hop Performance

Hop performance was assessed at 1-year post-ACLR on each leg using a single-forward hop, triple-crossover hop and side hop, as previously reported [24]. Detailed methodology for each hop test is available in a Supporting Information. Briefly, for the single-forward hop, participants held their hands behind their back and were instructed to hop as far forward as possible, landing with balance (no movement or shuffling of the foot). For the triple-crossover hop, the same instructions were used, with each participant asked to hop three times in succession, crossing two parallel lines on the floor 15 cm apart with each hop. For both forward hop tests, a minimum of three trials were completed on each leg, and the longest trial used in analysis [25, 26]. The side hop test measured the number of sideways (lateral) hops completed back and forth over two lines 40 cm apart [25]. Hands were placed on the hips throughout the test. Standardized encouragement was given for all tests. Raw scores were used in all analyses, as the use of metrics such as a limb symmetry index may overestimate physical capacity (e.g., symmetrical performance despite deficits in raw scores compared to normative data) [22]. Preliminary analyses showed that the single-forward hop and triple-crossover hop were highly correlated ($r = 0.90$, 95% CI 0.88–0.92), in line with previous work [7, 27]. As such, we chose to only use single-forward hop and side hop as predictor variables in our analyses.

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2.3.2 | Long-Term Outcomes: Self-Reported

Participants completed the KOOS at 1-, 3-, 5-, and 11-year post-ACLR. The KOOS is valid, reliable and measures self-reported perception of six different domains: pain, symptoms, sport and recreation, activities of daily living, and quality of life [28, 29]. For this study, we used the pain and quality of life subscales, as these are frequently impaired and relevant to this patient population [13, 15, 30].

2.3.3 | Subsequent Knee Events and Return-to-Sport

At each follow-up, participants were asked to report any injuries, surgeries or procedures sustained to either knee since their index ACLR, and the approximate timing. At the 3-year follow-up, this was completed online, and at 5- and 11 years follow-up information was recorded by a researcher in-person or via phone. We later classified a subsequent knee event as any new knee problem (e.g., knee effusion), injury (e.g., ACL rupture, meniscal tear) or surgery (e.g., arthroscopy), for the index or contralateral leg. The index limb was defined as the limb that underwent primary ACLR (i.e., first suffered ACL injury). We only used the first subsequent knee event in analysis, if a participant had multiple. We also asked participants whether they had returned to their preinjury level of sporting participation at all follow-ups, and, if yes, the approximate timing. Preinjury level of sport was defined as the same sport and level of competition prior to injury.

2.4 | Data Analysis

Statistical analyses were conducted in R Studio [31]. To investigate the trajectory of self-reported knee outcomes and explore the relationship between hop performance and future self-reported knee outcomes, we fit generalized linear mixed models using the "glmmTMB" package [32]. Traditional linear regression using ordinary least squares (OLS) is inappropriate for the distribution of data from patient-reported outcome scales such as the KOOS, as it assumes a continuous and unbounded response variable, which can lead to biased estimates and predictions outside the plausible range of the scale [33, 34]. Therefore, an "ordered beta distribution" was chosen [35] given all KOOS scales are anchored between 0 and 100 in a semi-ordinal scale, and inspection of distributions aligned best with a beta distribution. Scores were transformed to a beta distribution [0,1] by dividing the KOOS subscale score by 100. For our final models for each KOOS subscale (pain, QOL), we fit models with random effects for individuals (random intercept per participant, fixed slope shape) and predictor variables of hop performance (single-forward hop or side hop) and timepoint (years and quadratic polynomial). We controlled for covariates of age, BMI, sex, and concomitant injury (presence/absence of ligament or meniscal/cartilage injury accompanying ACL rupture).

The following process was used to arrive at our final models. To allow for multiple timepoints per participant, we trialed different model structures including random effects for individuals; or random effects for individuals and timepoints (i.e., random intercept and slope). We then trialed linear or polynomial terms for timepoint (to allow for potentially nonlinear trajectory [i.e., inverted U shape]). We also explored potential nonlinear association [36] between hop performance and KOOS subscales by incorporating restricted cubic splines. Spline knot positions were fitted using the "rms" package [37] as recommended by Harrel [38]. As all our independent variables were measured at baseline, we chose not to include a potentially time-varying dichotomous variable controlling for reinjury status. Final models were chosen based on model fit heuristics, such as Akaike information criterion (AIC), and visual inspection. The DHARMa package was used to ensure adequate model fit and performance (residual diagnostic) before reporting results [39]. We also tested for zero or one inflation (i.e., more frequent observations of 0 or 1 than expected) and found no significant evidence. We used "all available data" in our models (i.e., all cases, even if incomplete data at some follow-ups) and did not impute any missing values as this may introduce bias without benefits in performance, even with large dropout numbers [40].

Results of the estimated trajectory of self-reported knee outcomes over time (KOOS-pain and QOL subscale) are presented with plots of estimated marginal means from fitted models, alongside the original Likert scale of the KOOS to assist in interpretation [41]. To present findings for the relationship with hop performance, we report the average marginal effects (AME, the average percentage change in KOOS subscale with a 1 unit change in predictor variable [hop performance]) and plots for interpretation ("marginaleffects," "ggeffects," and "emmeans" packages). Full regression results and analysis scripts are available in a Supporting Information.

To measure the relationship between hop performance at 1 year and subsequent knee events, we conducted a survival analysis using Cox proportional hazards models. We used the "survival" package [42] using all available datapoints for each participant. Participants were censored at the last follow-up available if no subsequent knee event was reported. Timepoints were calculated as months between ACLR and the participant's assessment. We fit separate models for: (i) index events; and (ii) any events (index or contralateral side), with single-forward hop or side hop as predictor variables, controlling for age, sex, BMI, and concomitant injury. We also included a time-varying covariate of return to preinjury level of sport, given exposure to previous level of sport could impact risk of new injuries. Results are reported as hazard ratios (HR) and 95% confidence intervals (CIs).

3 | Results

We recruited 124 participants at 1-year post-ACLR (baseline demographics Table 1), with dropouts across all follow-ups detailed in Figure 1. There were no major differences in age, sex, BMI, or baseline KOOS scores between included participants and dropouts at 3 and 5 years; however, at 11 years more men had dropped out than women (odds ratio 3.55, 95% CI 1.59–8.50). 72 participants had 11-year follow-up information. Mean (SD, range) follow-up time at each timepoint was 1.1 (0.1, range = 0.9–1.2) years at 1-year follow-up; 3.0 (0.1, range = 2.8–3.5) years at 3-year follow-up; 5.2 (0.3,

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TABLE 1 | Baseline characteristics of all participants, and those who dropped out.

Variable	1-Year post-ACLR $n=124$	3-Year drop out $n=10$	5-Year drop out $n=35$	11-Year drop out $n=52$
Female, number (%)	43 (35%)	2 (20%)	11 (31%)	10 (19%)
Age, years	30.8 (8.7)	31.5 (9.8)	29.3 (7.5)	29.7 (7.3)
Height, m	1.76 (0.09)	1.78 (0.07)	1.74 (0.08)	1.77 (0.08)
Weight, kg	80.7 (15.8)	81.4 (14.5)	79.8 (16.9)	83.1 (16.4)
Waist circumference, cm	85.1 (11.2)	84.4 (10.0)	84.7 (12.2)	86.7 (11.3)
Concomitant injury/surgery, number (%)	62 (50%)	7 (70%)	15 (43%)	27 (52%)
Affected limb forward hop, cm	107.0 (30.2)	117.2 (24.5)	113.8 (32.0)	113.4 (29.5)
Contralateral limb forward hop, cm	120.4 (23.2)	123.1 (21.1)	123.7 (27.0)	124.2 (24.9)
Affected limb forward hop normalized (cm/height)	60.9 (19.9)	65.8 (13.1)	65.5 (18.4)	64.0 (16.8)
Contralateral limb forward hop normalized (cm/height)	68.5 (12.5)	69.1 (10.8)	71.1 (15.2)	70.1 (13.9)
Affected limb side hop, reps	26.7 (14.8)	33.6 (12.3)	29.9 (17.4)	28.6 (16.2)
Contralateral limb side hop, reps	32.4 (13.5)	37.1 (9.6)	34.1 (14.8)	33.4 (13.0)
KOOS-Symptoms	83.8 (13.0)	82.9 (14.1)	81.2 (14.6)	82.4 (11.7)
KOOS-Pain	91.0 (9.0)	91.4 (6.9)	92.3 (8.4)	91.7 (8.1)
KOOS-Activities of daily living	97.0 (5.7)	98.9 (1.9)	98.1 (2.7)	96.8 (6.8)
KOOS-Sport and Recreation	81.8 (16.2)	86.0 (1.9)	84.7 (15.1)	83.5 (15.4)
KOOS-Quality of life	68.4 (18.3)	68.1 (17.0)	70.4 (17.6)	68.5 (19.3)

Note: All values are mean (SD), unless otherwise denoted.
Abbreviation: KOOS, knee injury and osteoarthritis outcome score.

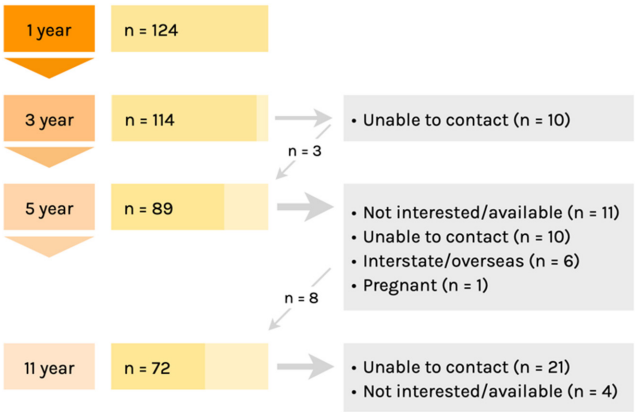

FIGURE 1 | Flow of participants through study, with gray box showing drop outs at each timepoint. Participants were re-contacted at each timepoint regardless of previous follow-up participation. Small diagonal arrows indicate participants who previously dropped out who were included in subsequent timepoints.

range = 4.8–6.2) years at 5-year follow-up and 11.5 (0.4, range = 10.7–12.2) years at 11-year follow-up. Over the 11 years since ACLR, there were 21 index secondary knee events (6 ACL re-ruptures, $n=117 \geq 3$ years follow-up, Table 2) and 11 contralateral secondary knee events (5 new contralateral ACL ruptures, Table 2).

TABLE 2 | Details of first subsequent knee event ($n=117$ with follow-up ≥ 3 years).

	Index knee	Contralateral knee
ACL rupture	6	5
Arthroscopy	7	2
Meniscal surgery	3	1
Meniscal injury	2	0
Collateral ligament injury	2	2
Cartilage injury	1	0
Knee effusion	0	2
Total	21	12

Note: $n=1$ sustained injury on index and contralateral knee.
Abbreviation: ACL, anterior cruciate ligament.

3.1 | Self-Reported Knee Outcomes

3.1.1 | KOOS-Pain

We found a polynomial relationship between timepoint and KOOS-pain, with a slight improvement at 3 and 5 years (2 and

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3 point mean improvement from 1 year) tailing off slightly by 11 years (1 point improvement from 1 year, Figure 2). There was a modest positive relationship between 1-year forward hop distance and KOOS-pain (Figure 3), with questionable clinical utility (AME (average percentage change in KOOS with a + 1 cm forward hop) = 0.06% (95% CI 0.02%–0.10%, $z = 2.85$, $p = 0.004$)). Importantly, there was major individual variation with the intraclass correlation coefficient (ICC) for individual-level random effects being 0.67 (Figure 2). This relationship was similar for side hop performance, with a very small relationship between 1 year greater baseline side hop performance and better KOOS-pain scores (AME = 0.10% (95% CI 0.02%–0.19%), $z = 2.30$, $p = 0.021$).

3.1.2 | KOOS-QOL

For KOOS-QOL, there was a similar polynomial relationship for timepoint (3-year follow-up = 6-point improvement, 5-year follow-up = 10-point improvement, 11-year follow-up = 6-point improvement from 1 year, Figure 4). We found a nonlinear (spline) relationship between single-forward hop distance and KOOS-QOL (Figure 5), with a potentially meaningful positive relationship for lower hop scores (< 108 cm AME = 0.36% (95% CI 0.19–0.52), $z = 4.26$, $p < 0.001$), but no significant relationship for higher hop scores (> 108 cm AME = 0.01% (95% CI −0.15 to 0.18), $z = 0.12$, $p = 0.90$). We also found a similar nonlinear (spline) relationship between side hop performance and KOOS-QOL with lower scores (< 25 reps) being related to worse QOL (AME = 0.62% (95% CI 0.23–1.02), $z = 3.10$, $p = 0.002$), but no significant relationship for higher scores (> 25 reps AME = 0.02% (95% CI −0.27 to 0.30), $z = 0.11$, $p = 0.92$). The ICC for individual-level random effects for both models (forward hop 0.35, side hop 0.38) indicated moderate individual-level variation not explained by fixed effects.

3.2 | Secondary Knee Events

Single-forward hop and side hop performance were not related to risk of a subsequent knee event on the index knee (single-forward hop HR = 0.99, 95% CI 0.98–1.02; side hop HR = 1.00, 95% CI 0.96–1.04). Hop performance was also not related to risk of any subsequent knee event (index or contralateral) (single-forward hop HR = 1.01, 95% CI 0.99–1.02; side hop HR = 1.02, 95% CI 0.99–1.05). None of the fitted models for either outcome showed any predictive value (no significant predictors or covariates, Supporting Information). The survival probability at final follow-up was 0.77 (95% CI 0.68–0.86) for index subsequent knee events and 0.65 (95% CI 0.55–0.77) for any subsequent knee event.

4 | Discussion

Self-reported knee outcomes were largely stable across our population, up to 11-year post-ACLR, with major variance within the sample. Hop performance may explain some variability in long-term self-reported knee outcomes, though the relationship appears modest and potentially nonlinear—poor performance was related to worse knee-related QOL; however, a higher hop score may not be meaningfully linked to better QOL. No relationship between hop performance and risk of a subsequent knee event was observed. Poor hop performance might be a useful indicator of groups at risk of poor long-term prognosis; however, it is unlikely to be prognostic at an individual level. It is also unclear whether intervening to improve functional performance would modify trajectory of symptoms.

Knee health status was largely stable across the 10 years of our study. There was a significant polynomial relationship for time for both KOOS-pain and KOOS-QOL (showing slight

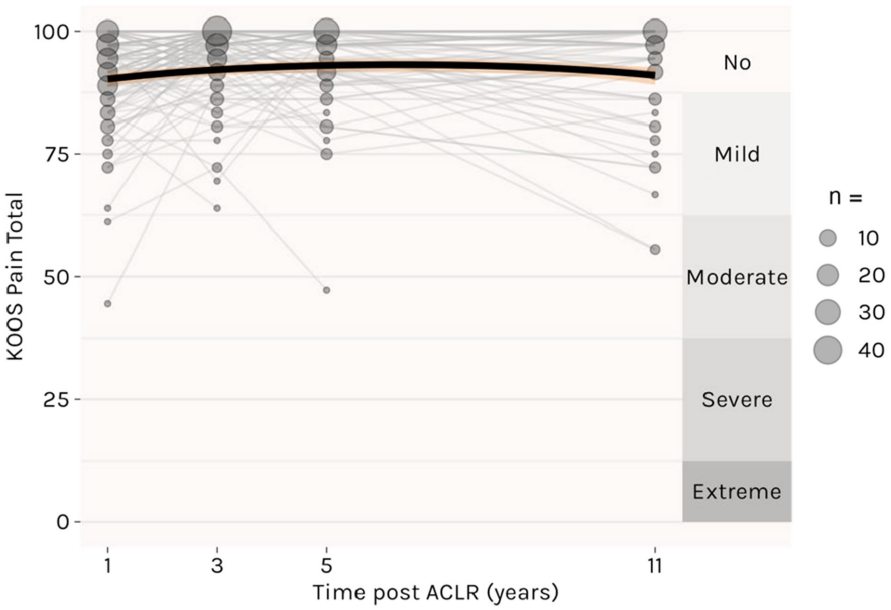

FIGURE 2 | Trajectory of KOOS-pain. Thick black line and orange shading show estimated marginal mean and 95% confidence interval from model fit. Point sizes indicate number of people (*n*) with faint gray lines individual trajectories. The categories on the right of the plot are based on the Likert scale used in the KOOS-pain are shown to simplify interpretation.

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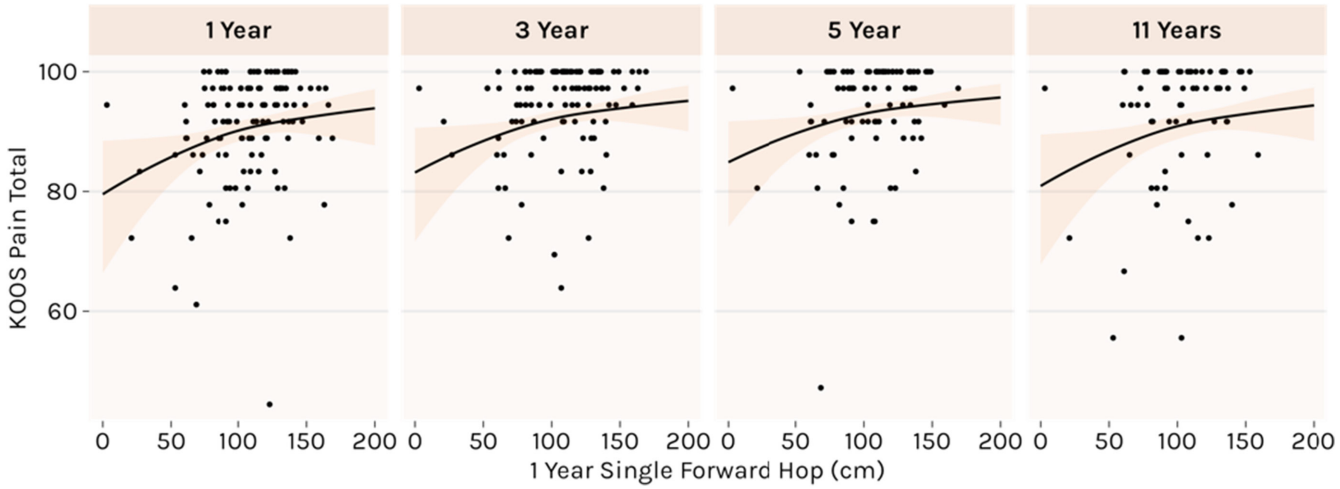

FIGURE 3 | Relationship between 1-year single-forward hop and KOOS-pain at each timepoint (controlling for age, sex, and BMI). Line and shading indicate estimated marginal mean and 95% confidence interval from model fit.

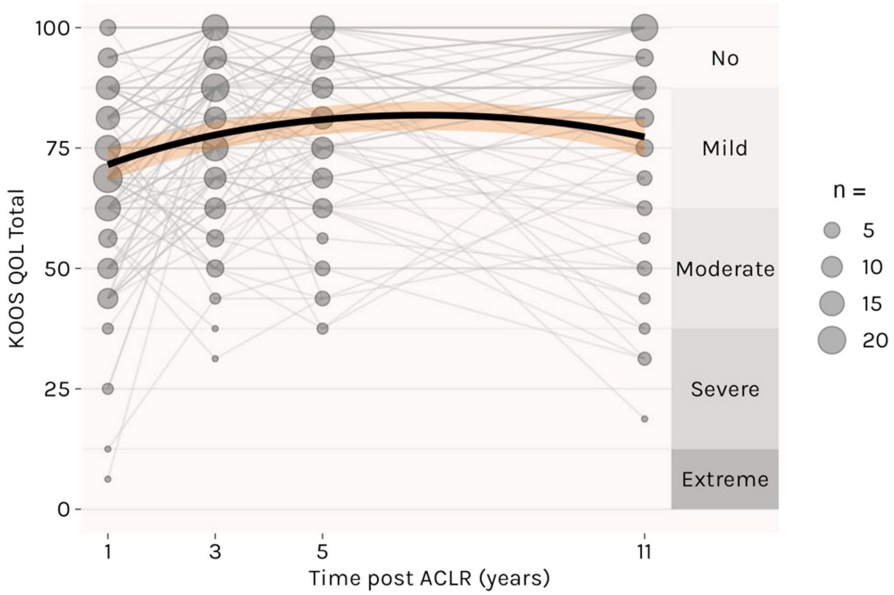

FIGURE 4 | Trajectory of KOOS-QOL. Thick black line and orange shading show estimated marginal mean and 95% confidence interval from model fit. Point sizes indicate number of people (*n*) with faint gray lines individual trajectories. The categories on the right of the plot are based on the Likert scale used in the KOOS-QOL are shown to simplify interpretation.

improvement at 3 and 5 years before beginning to worsen at 11 years); however, the predicted change was less than 5 points for KOOS-pain and 10 points for KOOS-QOL. These are less than measurement error (pain 6 points, QOL 7 points) and smallest detectable change (pain 16, QOL 18 points) for these subscales [30]. The stability in mean scores over time in our sample aligns with similarly aged knee-injured people, with more deterioration seen in older age (> 55 years) [43]. Based on the Likert scales used in the KOOS [41], on average our participants exhibited no difficulty with knee pain, and mildly impaired QOL. The KOOS-pain outcomes are similar to previous knee-injured cohorts [15], while QOL outcomes were lower than in some studies [15, 43]. Our cohort's mean subscale scores are also very similar to those reporting "acceptable symptoms" at 2-year post-ACLR [44]. Based on the trajectory in this study, knee health status at 1 year is likely to predict future knee health status. The major individual variability, not explained by predictor variables, highlights that distinct patient stories were not captured by our models.

Better hop performance may not be meaningfully associated with improved future knee pain and QOL. This aligns with previous systematic reviews of other medium-term follow-up studies [11, 12]. In our sample, based on predicted model fit, the range from worst to the best hop performer (for single-forward or side hop) was associated with a less than 10-point benefit on KOOS-pain across all timepoints. Though statistically significant, such differences are potentially clinically meaningless (i.e., not relevant for many patients) [30]. For KOOS-QOL, the relationship with hop performance was nonlinear, with no meaningful relationship for half of our population (i.e., hop score above median value). These findings, combined with the imprecision relating

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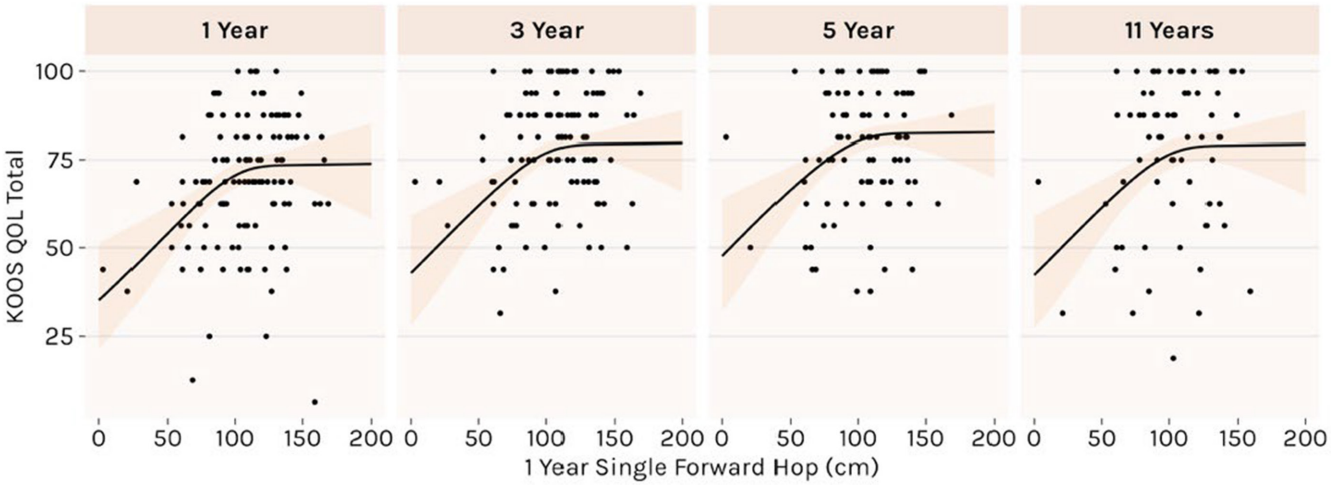

FIGURE 5 | Relationship between 1-year single-forward hop and KOOS-QOL at each timepoint (controlling for age, sex, and BMI). Line and shading indicate estimated marginal mean and 95% confidence interval from model fit.

to major individual-level variability reflected in the random effects fitted in our analyses, suggest that using hop performance to forecast individual prognosis is probably futile.

A nonlinear relationship between hop performance and future knee outcomes could explain some of the inconsistency seen in the literature regarding the prognostic value of functional tests [11]. Forward hop scores < 108 cm had a potentially meaningful link to knee-related QOL (10 cm greater single-forward hop = 3.5% better KOOS-QOL), which was not present for higher hop scores. Modeling these data with a linear relationship (the most used analysis in the literature) may have underestimated the overall effect [36]. The spline cutoff at 108 cm should not be used as a target or benchmark, and instead broadly indicates that worse hop performance (below the median in the current sample) may be more useful clinically for explaining prognosis. Hop tests are almost never measured in isolation though, and measuring self-reported knee outcomes such as the KOOS together with functional tests may be most informative to guide patient expectations and prognosis [45].

The rate of secondary knee events and secondary ACL rupture was similar or slightly lower in our population to previously reported. Our index ACL re-rupture (i.e., graft failure) rate of 5% and contralateral ACL rupture rate of 4%, was slightly lower than meta-analysis estimates of 7% and 8%, respectively [46]. Data on other secondary knee events, capturing other knee injuries (noncruciate ligament tears) or further treatment (e.g., meniscal surgery) are less commonly reported. Meniscal tears are most tracked, with secondary incidence of 7%–17% reported (with follow-up as long as 18 years post-ACLR), which is consistent with our study (11% reporting subsequent index meniscal/cartilage injury or arthroscopic surgery). As many as one in four people could be expected to have a secondary knee event in the 10 years after their ACLR. In addition to educating patients about the risk of graft failure and contralateral ACL tears, clinicians should also advise patient expectations on the potential likelihood of other subsequent knee injuries and procedures appropriately. Further work is also needed to determine what modifiable factors might be related to secondary injuries such as meniscal tears, and whether clinicians are able to decrease this risk through rehabilitation interventions.

Hop performance was not associated with risk of a subsequent knee event. Though our definition of a knee event was slightly broader than previous studies (which often focus on secondary ACL rupture), our findings support a recent meta-analysis showing no difference in risk of subsequent injury when achieving a common benchmark (90%) of limb symmetry on forward hop tests [11]. However, the literature in this space is conflicting—achieving symmetry criteria on functional tests is associated with higher and lower risk of reinjury [47, 48]. This inconsistency is potentially explained by whether sports participation was controlled for, given this is a key exposure factor [11]. We found returning to preinjury level of sport was not associated with risk of a subsequent knee injury. As our study was observational (i.e., rehabilitation not controlled for) return-to-sport testing was not routine, and we were unable to measure the impact of passing return-to-sport criteria on reinjury risk. Measuring hop performance also does not fully consider movement biomechanics, which could also contribute to reinjury risk. Interestingly, other known risk factors for secondary ACL injury including younger age [49], lower BMI, and male sex [50] were not confirmed in our population, which could be explained by our slightly older sample (mean age 31 years).

For clinicians, measuring hop performance continues to be valuable to determine readiness to return-to-sport and guide future rehabilitation goals [10]. Hop performance may have limited value in educating patients about their prognosis, with results only predictive at a group, but not individual level. Individual variability in outcome trajectory after ACLR and baseline status (i.e., their current self-reported knee health) is likely to trump functional performance in predicting future knee-related health after the first 12 months. Given self-reported knee status was stable on average after 1 year, this may be a better guide to long-term prognosis for patients. Importantly, it remains unclear whether intervening to improve poor hop performance at 1 year will lead to better future outcomes. Future clinical trials investigating interventions

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targeting those at risk of poor outcomes and early osteoarthritis are a priority [10] and will help to answer these questions [51]. Hop performance in this study was measured purely as physical performance, but assessing other aspects including knee confidence [52] and hop quality (i.e., biomechanics) may also be worthwhile. Recent studies have also suggested other tests such as the single leg vertical jump may be better at identifying performance deficits and aberrant knee biomechanics than forward hop tests [53, 54].

Our study had some limitations that should be considered. By the 11-year follow-up approximately 40% of participants had dropped out of the study, with potential attrition bias leading to overly optimistic knee health outcomes and failure to capture some subsequent knee events. Conversely, those with high knee function may not have seen the value in continuing participation at later follow-ups. Characteristics of those who dropped out were generally balanced at all follow-ups, though more men dropped out at 11 years. We made use of available data points and decided not to impute any missing data, so as not to introduce any additional bias, as recommended [40]. The high level of individual variability, as seen in random effects for individuals, highlights that key predictor variables may not have been accounted for in our models. We attempted to control for many known factors influencing self-reported knee outcomes and functional testing (i.e., age, sex, BMI, concomitant injury); however, other factors, such as patient expectations, as well as psychological [52], and mental health [55], may also need to be considered when estimating long-term knee health prognosis.

For our survival analysis, participant recall may have affected the accuracy of the timing and exact diagnosis of subsequent knee events. Errors in timing would likely only slightly have affected our estimates (e.g., if participant estimate inaccurate by 1 year), and we did not conduct any analyses on specific diagnoses. We controlled for whether participants had returned to preinjury level of sports competition, which potentially underestimates exposure given people may have changed sports (but possibly equally high risk), or continued participating at a lower level. Another limitation was only measuring three hop tests in two planes, as others, such as the single leg vertical jump, may better highlight movement deficits post-ACLR [53]. Lastly, with a larger sample size, we may have been able to investigate whether distinct trajectories exist after ACLR, with analyses such as latent growth mixture models or other similar approaches [56]. Future research with larger or pooled datasets, may be able to highlight important subgroups.

5 | Perspective

Hop performance at 1-year post-ACLR is related to self-reported knee pain and QOL up to 11 years. This relationship is not likely to be clinically meaningful, as the association was weak and individual variability was more influential on long-term prognosis. Groups with especially poor single-forward hop or side hop performance (lowest 10–20th percentile) may be at risk of worse future knee-related QOL. Hop performance (on any test) was not related to risk of a subsequent knee event after ACLR, (new injuries or subsequent surgery). Clinicians can continue to use hop testing to guide rehabilitation goals and decision making; however, it is less likely to be valuable for forecasting future outcomes.

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Ethics Statement

Ethical approval was obtained from La Trobe University (HEC20529), and all participants provided informed consent.

Conflicts of Interest

Kay Crossley holds a research grant from Levin Health. Timothy Whitehead has received funds as part of a clinical fellowship from Smith and Nephew and is a consultant to Medacta. He also has received payment from Smith and Nephew, Medacta and Anthrex for speaking engagements. Hayden Morris has received funds as part of a clinical fellowship from Oceania Orthopedics.

Data Availability Statement

Data are available on request. All analysis scripts are publicly available.

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